[METRIC]
H-B-420L
February 12, 1997
Superseding
H-B-420K
August 30, 1991

FEDERAL SPECIFICATION

BRUSH, PAINT

The General Services Administration has authorized the use of this Federal Specification, for all federal agencies

1. **SCOPE AND CLASSIFICATION.**

1.1 **SCOPE.** This specification covers flat, metal-bound, paint and varnish brushes made with natural hog bristle or synthetic filament. Master grade brushes are designed and manufactured for paint applications where superior paint loading and release are required. High grade brushes are for varnish and enamel applications. Medium grade brushes are excellent as sash and trim tools. Utility grade brushes deliver excellent results in general purpose painting operations where brush strokes are not a consideration.

1.2 **CLASSIFICATION.** The brushes shall be of the following types, grades, and sizes as specified (see 6.2): (Size denotes the width of brush inside the ferrule.)

	TYPE I - BRISTLE Size, mm (inches)		TYPE II - FILAMENT Size mm, (inches)	
Grade AA, Master Grade	152	(6)	127	(5)
	127	(5)	102	(4)
	102	(4)	89	(3-1/2)
	89	(3-1/2)	76	(3)
Grade A, High Grade	152	(6)	152	(6)
	127	(5)	127	(5)
	102	(4)	102	(4)
	89	(3-1/2)	89	(3-1/2)
	76	(3)	76	(3)
	64	(2-1/2)	64	(2-1/2)
	51	(2)	51	(2)
	38	(1-1/2)	38	(1-1/2)
	25	(1)	25	(1)
Grade B, Medium grade,	127	(5)	127	(5)
	102	(4)	102	(4)
	89	(3-1/2)	89	(3-1/2)
	76	(3)	76	(3)
	64	(2-1/2)	64	(2-1/2)

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any other data which may improve this document should be sent by letter to: Paints and Chemicals Center, Attn: 10FTE, 400 15th Street, S.W., Auburn, WA 98001.

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	TYPE I - BRISTLE Size, mm (inches)		TYPE II - FILAMENT Size mm, (inches)	
Grade B, Medium Grade	51	(2)	51	(2)
	38	(1-1/2)	38	(1-1/2)
	25	(1)	25	(1)
Grade C, Utility grade,	102	(4)	102	(4)
	89	(3-1/2)	89	(3-1/2)
	76	(3)	76	(3)
	64	(2-1/2)	64	(2-1/2)
	51	(2)	51	(2)
	38	(1-1/2)	38	(1-1/2)
	25	(1)	25	(1)
	13	(1/2)		----

2. **APPLICABLE DOCUMENTS**.

2.1 The following documents, of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

H-S-951 Synthetic filament for brushes, General Specification for
PPP-B-1880 Brushes; Preparation for Delivery of

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available from the GSA Specification Section (3FBP-W), Suite 8100, 470 L'Enfant Plaza, S.W., Washington, DC 20407).

2.2 **Other publications**. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Association, Inc., Traffic Department, 2200 Mill Road, Alexandria, VA 22314).

American Society For Quality Control

ANSI/ASQC Z1.4-1993 Sampling Procedures and Tables for Inspection by Attributes

American Society for Testing and Materials:

ASTM D13 Specification for Spirits of Turpentine

(Application for copies of ASTM Standards should be addressed to the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959).

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(Copies of ASQC Standards may be obtained from the American Society for Quality Control, 611 East Wisconsin Avenue, Milwaukee, WI 53202.)

3. **REQUIREMENTS.**

3.1 **Materials and components**. Materials and components shall conform to the requirements specified herein.

3.1.1 **Bristle (Type I)**. The bristle for Type I brushes shall be unbleached and undyed natural hog bristle. The bristle shall be boiled or steamed and straightened so that it will not twist or curl, and shall be free from reconditioned bristle or adulterants. The bristle shall be certified by the contractor to be free from added color or dyes, and shall pass the straightness test specified in 4.4.4. A certificate of compliance from the supplier may be accepted that the bristle meets the stiffness requirements.

3.1.2 **Synthetic filaments (Type II)**. The synthetic filaments for Type II brushes shall be an organic polymeric material conforming to H-S-951, class N, type I, or class E, type I tapered filaments meeting the size designation of Table II. All filaments shall be tipped to a depth of not less than 5 mm. (3/16 inch). After tipping, 20 percent, by weight of the top-sized filaments, shall be flagged. (See 6.3.1).

3.1.3 **Ferrule**. The ferrule shall be made of corrosion resistant metal, or metal made corrosion resistant by electro-plating. The coating shall not wrinkle, blister, separate, or become soft or dull when tested in accordance with 4.4.2.2.

3.1.4 **Handle**. The handle shall be either a smoothly finished close grained hardwood, or a synthetic material with all sprues and runners removed. The handle shall not crack or break when tested as in 4.4.3. When tested in accordance with 4.4.2.1, the handle coating shall not become softened or dull.

3.1.5 **Filler strips**. Filler strips shall be made of wood, metal, or plastic, and shall not soften or otherwise change when tested as in 4.4.2.2.

3.1.6 **Setting Compound**. The setting compound shall be a material which shall withstand the test specified in 4.4.2.2 without softening or deteriorating, and without loosening of bristle or filament in the compound. No more than a total of five bristles or filaments shall be removed during the pull test specified in 4.4.2.2(c).

3.2 **Construction and Physical Requirements.**

3.2.1 **Ferrule**. The ferrule metal shall have a minimum wall thickness of 3 mm (0.010 inch). The ferrule shall be formed with an interlocked, lock-seamed, soldered, or welded end lap, and shall have a groove at the base to aid in controlling the flow of the setting compound. The ferrule height shall be a minimum of 32 mm (1-1/4 inch) for grade C brushes and 37 mm (1-15/32 inch) minimum for grade AA, A and B brushes.

3.2.2 **Set**. The butt ends of the bristles or filaments shall be set in a setting compound conforming to 3.1.6. When tested, as in 4.4.2.2, there shall be no more than a total of five (5) bristles or filaments pulled loose during the tensile pull.

3.2.3 **Assembly**. For grades AA and A brushes, the ferrule shall be fastened to the handle with 16 or 17 gage, corrosion resistant nails. The minimum length of the nails shall be such that the shank of the nail shall engage at least 1/2 the thickness of the brush. There shall be at least two (2) nails on each side of brushes 76 mm (3 inches) wide or smaller; three (3) nails on each side of the 89 mm. (3-1/2 inch) and 102 mm. (4 inch) brushes; and four (4) nails on each side of the 127 mm. (5 inch) and 152 mm. (6 inch) brushes. Nails shall be evenly spaced on each side of the brush. For grades B and C, the ferrules may be nailed or otherwise securely fastened to the handles. If the ferrules are fastened by a means other than nailing, the assembled brushes shall be dropped as specified in 4.4.3, and shall show no separation or loosening of the ferrule to handle joint.

3.2.4 **Trim**.

a) All grade AA brushes shall be formed to a square top with flagged ends preserved, or cup chiseled, with a minimum chisel of 6 mm (1/4 inch).

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b) Grade A, and grade B brushes (wider than 76 mm (3 inches) and all grade C brushes shall be formed to a square top, with tipping and flagging preserved.
c) 76 mm (3 inch) brushes shall be formed to a square top (with the tipped or flagged ends preserved), or cupped or machined to a chisel edge with a minimum chisel of 6 mm (1/4 inch). (If cup chiseled, flag ends shall be preserved. If machine chiseled, the ends shall be sanded to a fine, soft edge).

d) Grade A and grade B brushes ((less than 76 mm (3 inches) wide)) shall be cupped or machined to a chisel edge with a minimum chisel of 6 mm (1/4 inch). If cup chiseled, flagged ends shall be preserved. If machine chiseled, the ends shall be sanded to a fine, soft edge.

3.3 **Composition and Physical Requirements**.

3.3.1 **Type I.** The bristle shall be 60 percent top size and blended in accordance with Table I. (Dressed bristle, described as 60 percent top size, denotes that bundle consists of 60 percent bristle measuring 2 mm. (1/16 inch) or more above the length specified to 3 mm (1/8 inch) below the length specified; 30 percent to consist of bristle from 3 mm (1/8 inch) to 10 mm (3/8 inch) below the length specified; 10 percent will be 10 mm (3/8 inch) to not more than 13 mm (1/2 inch) below specified length, but including a few unavoidable shorts).

TABLE I. Blend of bristle and physical requirements for type I brushes

Grade	Brush width inside ferrule, mm., (in), +/_ 1 mm (1/32 in.)	Percentage (wt.) & length of bristle mm., (in), Min.	Brush thickness inside ferrule mm, (in), +/- 1 mm (1/32 in.)	Bristle length outside ferrule mm, (in), Min.	Bristle weight per brush, gm, Min.
AA	152 (6) [square or cup	80% 133 (5-1/4) 10% 127 (5) 10% 114 (4-1/2)	25 (1)	119 (4-11/16)	153
AA	127 (5) [square or cup]	80% 133 (5-1/4) 10% 127 (5) 5% 121 (4-3/4) 5% 114 (4-1/2)	25 (1)	127 (5)	128
AA	102 (4) [square or cup]	80% 127 (5) 10% 121 (4-3/4) 5% 114 (4-1/2) 5% 108 (4-1/4)	25 (1)	113 (4-7/16)	100
AA	89 (3.5) [square or cup]	80% 121 (4-3/4) 10% 114 (4-1/2) 5% 108 (4-1/4) 5% 102 (4)	25 (1)	106 (4-3/16)	82
A	152 (6) [square]	80% 121 (4-3/4) 10% 114 (4-1/2) 5% 108 (4-1/4) 5% 102 (4)	25 (1)	106 (4-3/16)	125
A	127 (5) [square]	80% 121 (4-3/4) 10% 114 (4-1/2) 5% 108 (4-1/4) 5% 102 (4)	25 (1)	106 (4-3/16)	105
A	102 (4) [square]	80% 114 (4-1/2) 10% 108 (4-1/4) 5% 102 (4) 5% 95 (3-3/4)	25 (1)	100 (3-15/16)	79
A	89 (3.5) [square]	80% 114 (4-1/2) 10% 102 (4) 5% 95 (3-3/4) 5% 89 (3-1/2)	25 (1)	100 (3-15/16)	65
A	76 (3) [square]	80% 95 (3-3/4) 15% 89 (3-1/2) 5% 83 (3-1/4)	21 (13/16)	78 (3-1/16)	60

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TABLE I. Blend of bristle and physical requirements for type I brushes

Grade	Brush width inside ferrule, mm., (in), +/_ 1 mm (1/32 in.)	Percentage (wt.) & length of bristle mm., (in), Min.	Brush thickness inside ferrule mm, (in), +/- 1 mm (1/32 in.)	Bristle length outside ferrule mm, (in), Min.	Bristle weight per brush, gm, Min.
A	64 (2.5) [chisel]	80% 89 (3-1/2) 15% 83 (3-1/4) 5% 76 (3)	21 (13/16)	71 (2-13/16)	40
A	51 (2) [chisel]	80% 83 (3-1/4) 15% 76 (3) 5% 70 (2-3/4)	19 (3/4)	65 (2-9/16)	27
A	38 (1.5) [chisel]	80% 83 (3-1/4) 15% 76 (3) 5% 70 2-3/4)	18 (11/16)	59 (2-5/16)	16
B	127 (5) [square]	80% 114 (4-1/2) 10% 108 (4-1/4) 5% 102 (4) 5% 95 (3-3/4)	22 (7/8)	100 (3-15/16)	96
B	102 (4) [square]	80% 102 (4) 10% 95 (3-3/4) 5% 89 (3-1/2) 5% 83 (3-1/4)	22 (7/8)	87 (3-7/16)	68
B	89 (3.5) [square]	80% 95 (3-3/4) 10% 89 (3-1/2) 5% 83 (3-1/4) 5% 76 (3)	22 (7/8)	81 (3-3/16)	57
B	76 (3) [square]	80% 89 (3-1/2) 10% 83 (3-1/4) 5% 76 (3) 5% 70 (2-3/4)	18 (11/16)	75 (2-15/16)	45
B	76 (3) [chisel]	80% 83 (3-1/4) 15% 76 (3) 5% 70 (2-3/4)	18 (11/16)	65 (2-9/16)	40
B	64 (2.5) [chisel]	80% 83 (3-1/4) 15% 76 (3) 5% 70 (2-3/4)	16 (5/8)	59 (2-5/16)	27
B	51 (2) [chisel]	100% 70 (2-3/4)	14 (9/16)	53 (2-1/16)	17
B	38 (1.5) [chisel]	100% 64 (2-1/2)	13 (1/2)	46 (1-13/16)	11
B	25 (1) [chisel]	100% 64 (2-1/2)	11 (7/16)	46 (1-13/16)	6
C	102 (4) [square]	80% 89 (3-1/2) 10% 83 (3-1/4) 5% 76 (3) 5% 70 (2-3/4)	22 (7/8)	75 (2-15/16)	54
C	89 (3.5) [square]	80% 83 (3-1/4) 10% 76 (3) 5% 70 (2-3/4) 5% 64 (2-1/2)	22 (7/8)	68 (2-11/16)	45
C	76 (3) [square]	80% 76 (3) 10% 70 (2-3/4) 5% 64 (2-1/2) 5% 57 (2-1/4)	18 (11/16)	62 (2-7/16)	37
C	76 [square]	100% 70 (2-3/4)	13 (1/2)	49 (1-15/16)	28
C	51 (2) [square]	100% 51 (2)	8 (5/16)	33 (1-5/16)	5

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TABLE I. Blend of bristle and physical requirements for type I brushes

Grade	Brush width inside ferrule, mm., (in), +/_ 1 mm (1/32 in.)	Percentage (wt.) & length of bristle mm., (in), Min.	Brush thickness inside ferrule mm, (in), +/- 1 mm (1/32 in.)	Bristle length outside ferrule mm, (in), Min.	Bristle weight per brush, gm, Min.
C	40 (1.5) [square]	100% 64 (2-1/2)	10 (13/32)	33 (1-5/16)	10
C	25 (1) [square]	100% 64 (2-1/2)	8 (5/16)	33 (1-5/16)	3
C	0.5 (13) [square]	100% 51 (2)	6 (1/4)	33 (1-5/16)	2

Bristles 133 mm (5-1/4 inch) through 127 mm (5 inch) shall be 10% soft, 20% medium, and 79% stiff. Bristles 121 mm (4-3/4 inch) through 114 mm (4-1/2 inch) shall be 10% soft, 20% medium, and 70% stiff. Bristles 108 mm (4-1/4 inch) through 102 mm (4 inch) shall be 25% soft, 30% medium and 45% stiff. Bristles 95 mm (3-3/4 inch) shall be 40% soft, 30% medium, and 30% stiff. Bristles 89 mm (3-1/2 inch) shall be 50% soft, 25% medium, and 25% stiff. Bristles 83 mm (3-1/4 inch) through 76 mm (3 inch) shall be 80% soft and 20% medium. Bristles shorter than 76 mm (3 inches) shall be 100% soft. (A tolerance of plus or minus 3% is allowed for each length and stiffness specified.)

3.3.2 **Type II.** Synthetic filament conforming to 3.1.2 shall be blended as specified in table II, with a tolerance of plus or minus 3 percent for each length and stiffness specified.

TABLE II. Blend of filament and physical requirements for type II brushes

Grade	Brush width inside ferrule, mm, in., +/- 1 mm (1/32 in.) [Trim]	Filament size designation, percentage, and length mm (inches), min.	Length of filament outside ferrule mm, (in.), min	Brush thickness inside ferrule mm, (in.), min.	Filament weight per brush gm., min.
AA	127 (5) [square or chisel]	**12-8:** 40% 121 (4-3/4) **15-10:** 15% 121 (4-3/4) 15% 114 (4-1/2) 15% 108 (4-1/4) 15% 102 (4)	106 (4-3/16)	25 (1)	130
AA	102 (4) [square or chisel]	**12-8:** 40% 121 (4-3/4) **15-10:** 15% 121 (4-3/4) 15% 114 (4-1/2) 15% 108 (4-1/4) 15% 102 (4)	106 (4-3/16)	25 (1)	105
AA	89 (3.5) [square or chisel	**12-8:** 45% 114 (4-1/2) **15-10:** 10% 114 (4-1/2) 15% 108 (4-1/4) 15% 102 (4) 15% 95 (3-3/4)	100 (3-15/16)	25 (1)	91
AA	76 (3) [square or chisel]	**12-8:** 55% 114 (4-1/2) **15-10:** 15% 102 (4) 15% 95 (3-3/4) 15% 89 (3-1/2)	98 (3-13/16)	25 (1)	74

TABLE II. Blend of filament and physical requirements for type II brushes

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Grade	Brush width inside ferrule, mm, in., +/- 1 mm (1/32 in.) [Trim]	Filament size designation, percentage, and length mm (inches), min.	Length of filament outside ferrule mm, (in.), min	Brush thickness inside ferrule mm, (in.), min.	Filament weight per brush gm., min.
A	152 (6) [square]	**12-8:** 45% 114 (4-1/2) **15-10:** 10% 114 (4-1/2) 15% 108 (4-1/4) 15% 102 (4) 15% 95 (3-3/4)	100 (3-15/16)	25 (1)	145
A	127 (5) [square]	**12-8:** 45% 114 (4-1/2) **15-10:** 10% 114 (4-1/2) 15% 108 (4-1/4) 15% 102 (4) 15% 95 (3-3/4)	100 (3-15/16)	25 (1)	116
A	102 (4) [square]	**12-8:** 55% 108 (4-1/4) **15-10:** 15% 102 (4) 15% 95 (3-3/4) 15% 89 (3-1/2)	94 (3-11/16)	25 (1)	88
A	89 (3.5) [square]	**12-8:** 55% 102 (4) 15% 95 (3-3/4) **15-10:** 15% 89 (3-1/2) 15% 83 (3-1/4)	87 (3-7/16)	25 (1)	77
A	76 (3) [square]	**12-8:** 55% 95 (3-3/4) 15% 89 (3-1/2) 15% 83 (3-1/4) **15-10:** 15% 76 (3)	81 (3-3/16)	25 (1)	68
A	76 (3) [chisel]	**12-8:** 40% 95 (3-3/4) 20% 89 (3-1/2) 20% 83 (3-1/4) **9-5:** 20% 95 (3-3/4)	78 (3-1/16)	21 (13/16)	54
A	64 (2.5) [chisel]	**12-8:** 20% 89 (3-1/2) 20% 83 (3-1/4) 20% 76 (3) **9-5:** 40% 89 (3-1/2)	71 (2-13/16)	21 (13/16)	45
A	51 (2) [chisel]	**12-8:** 10% 83 (3-1/4) 20% 76 (3) 20% 70 (2-3/4) **9-5:** 50% 83 (3-1/4)	65 (2-9/16)	19 (3/4)	35

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TABLE II. Blend of filament and physical requirements for type II brushes

Grade	Brush width inside ferrule, mm, in., +/- 1 mm (1/32 in.) [Trim]	Filament size designation, percentage, and length mm (inches), min.	Length of filament outside ferrule mm, (in.), min	Brush thickness inside ferrule mm, (in.), min.	Filament weight per brush gm., min.
A	38 (1.5) [chisel]	**12-8:** 20% 70 (2-3/4) 20% 64 (2-1/2) **9-5:** 60% 76 (3)	59 (2-5/16)	18 (11/16)	18
B	127 (5) [square]	**12-8:** 45% 114 (4-1/2) **15-10:** 10% 114 (4-1/2) 15% 108 (4-1/4) 15% 102 (4) 15% 95 (3-3/4)	98 (3-7/8)	22 (7/8)	102
B	102 (4) [square]	**12-8:** 55% 102 (4) 15% 95 (3-3/4) **15-10:** 15% 89 (3-1/2) 15% 83 (3-1/4)	86 (3-3/8)	22 (7/8)	74
B	89 (3.5) [square]	**12-8:** 55% 95 (3-3/4) 15% 89 (3-1/2) 15% 83 (3-1/4) **15-10:** 15% 76 (3)	79 (3-1/8)	22 (7/8)	60
B	76 (3) [square]	**12-8:** 55% 89 (3-1/2) 15% 83 (3-1/4) 15% 76 (3) 15% 70 (2-3/4)	73 (2-7/8)	22 (7/8)	48
B	76 (3) [chisel]	**12-8:** 20% 70 (2-3/4) 20% 76 (3) 10% 83 (3-1/4) **9-5:** 50% 83 (3-1/4)	49 (1-15/16)	18 (11/16)	42
B	64 (2.5) [chisel]	**12-8:** 20% 64 (2-1/2) 20% 70 (2-3/4) **9-5:** 60% 76 (3)	59 (2-5/16)	16 (5/8)	30
	51 (2) [chisel]	**12-8:** 30% 64 (2-1/2) **9-5:** 70% 70 (2-3/4)	52 (2-1/16)	14 (9/16)	20
B	38 (1.5) [chisel]	**9-5:** 30% 57 (2-1/4) 70% 64 (2-1/2)	46 (1-13/16)	13 (1/2)	13
B	25 (1) [chisel]	**9-5:** 30% 57 (2-1/4) 70% 64 (2-1/2)	46 (1-13/16)	11 (7/16)	6

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TABLE II. Blend of filament and physical requirements for type II brushes

Grade	Brush width inside ferrule, mm, in., +/- 1 mm (1/32 in.) [Trim]	Filament size designation, percentage, and length mm (inches), min.	Length of filament outside ferrule mm, (in.), min	Brush thickness inside ferrule mm, (in.), min.	Filament weight per brush gm., min.
C	102 (4) [square]	**12-8:** 55% 89 (3-1/2) 15% 83 (3-1/4) 15% 76 (3) 15% 70 (2-3/4)	75 (2-15/16)	22 (7/8)	54
C	89 (3.5) [square]	**9-5:** 25% 83 (3-1/4) **12-8:** 30% 83 (3-1/4) 15% 76 (3) 15% 70 (2-3/4) 15% 64 (2-1/2)	68 (2-11/16)	22 (7/8)	45
C	76 (3) [square]	9-5: 40% 76 (3) 12-8: 20% 76 (3) 20% 70 (2-3/4) 20% 64 (2-1/2)	62 (2-7/16)	22 (7/8)	34
C	76 (3) [square]	**9-5:** 100% 76 (3)	59 (2-5/16)	13 (1/2)	31
C	64 (2.5) [square]	**9-5:** 100% 70 (2-3/4)	52 (2-1/16)	12 (15/32)	21
C	51 (2) [square]	**9-5:** 100% 70 (2-3/4)	52 (2-1/16)	11 (7/16)	16
C	38 (1.5) [square]	**9-5:** 100% 64 (2-1/2)	46 (1-13/16)	10 (13/32)	10
C	25 (1) [square]	**9-5:** 100% 64 (2-1/2)	46 (1-13/16)	9.5 (3/8)	6
C	13 (0.5) [square]	**9-5:** 100% 64 (2-1/2)	33 (1-5/16)	6 (1/4)	2.75

3.4 **Marking for identification**. Unless otherwise specified (6.2), and in addition to any special marking required by the contract or order, each brush shall be stamped or otherwise permanently marked with the manufacturer's name or trademark and the National Stock Number (NSN).

3.5 **Workmanship**. The finished brushes shall be exempt of sharp edges, splinters, burrs, or other imperfections. In addition, the brushes shall be free of residual debris and other extraneous matter. The brushing part shall be trimmed evenly and uniformly with the flagged ends preserved to the finished length outside the ferrule. The brushes shall be well made and free from defects that will affect serviceability, durability or appearance. The brushing part must not exceed the flare measurement criteria and tolerances..

4. QUALITY ASSURANCE PROVISIONS.

4.1 **Responsibility for inspection**. Unless otherwise specified in the contract or order, the supplier is responsible for the performance of all inspection requirements specified herein using facilities acceptable to the Government. The Government reserves the right to perform any inspection herein when deemed necessary to assure that the brushes conform to prescribed requirements.

4.2 **Inspection of preparation for delivery**. An examination shall be made to determine compliance with 3.3 and section 5 of this specification. The examination shall be conducted as specified in PPP-B-1880.

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4.3 **Inspection of the end item.** Lots shall be assembled and randomly sampled in accordance with ANSI/ASQC Z1.4-1993, inspection level II, with an acceptable quality level (AQL) of 2.5 percent defective. Sample brushes shall be examined for the following attributes:

Bristles or filaments
Ferrules
Handles
Construction
Setting Compound
Trim
Workmanship
Identification Markings

Any brush in the sample containing one or more visual or dimensional defects shall be considered a defective brush.

4.3.1 **Sampling for testing of the end item.** A random sample of brushes shall be selected from each lot in accordance with ANSI/ASQC Z1.4-1993, inspection level S-2, with an acceptable quality level (AQL) of 2.5 percent defective.

4.4 **Testing procedures.** Samples, selected in accordance with 4.3.1, shall be subjected to the following tests. Failure to pass any test shall be cause for rejection of the lot. All tests shall be evaluated for conformance to the requirements in section 3.

4.4.1 **Weight of bristles or filaments.** After removal of the ferrule, the setting compound shall be removed from the bristles or filaments by any practical means which does not injure the bristles or filaments. The weight of bristles or filaments shall be determined for compliance with physical requirements.

4.4.2 **Solvent Resistance.** Solvent resistance tests shall be performed at room temperature.

4.4.2.1 **Handle (Organic Coating).** Brush handles with organic coating shall be tested as follows:

A. With the brushing end up, approximately one-half of the handle shall be immersed for 30 minutes in a commercially available mineral spirits. After 30 minutes, the handle shall be removed and examined. There shall be no wrinkling, blistering, or separation of the coating.

B. With the brushing end up, approximately one-half of the handle shall be immersed for 30 minutes in turpentine conforming to ASTM D13. After 30 minutes, the handle shall be removed from the turpentine and examined. There shall be no wrinkling, blistering, or separation of the coating.

C. The brush handle shall be dried for 2 hours and the organic coating shall be examined. The coating shall not be wrinkled, blistered, or separated from the handle and the coating shall not be soft.

4.4.2.2 **Base material, filler strips, synthetic handles**. The brush shall be tested for solvent resistance using a commercially available lacquer thinner. (NOTE: The organic coating on the handle may be removed by this solvent. The objective is to test solvent resistance of the handle base material, filler strips, and the setting compound.)

The brush shall be suspended so that the brushing part and approximately one-half of the ferrule are immersed in commercial lacquer thinner for 1 hour. Synthetic handles shall be immersed to a depth of one-half of the handle length. The brush shall be removed and examined as follows:

A. **Handle**. As soon as the brush is removed from the lacquer thinner, the brush shall be dried for 2 hours and examined for deterioration of the handle base material.

B. **Filler strips**. Filler strips shall be examined for softening, swelling, or other deleterious change.

C. **Setting compound**. The setting compound shall be examined for softening, blistering, cracking, or crazing. After drying the brush, the brush shall be suspended by the handle in a holding device, and a group of bristles, 6 mm. (1/4 in.) in diameter, shall be given a tensile pull of 46 N (10 lb.) away from the ferrule. Repeat this tensile pull test at random places one time for each inch of brush width.

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4.4.3 **Drop test**. Remove bristle (or synthetic filament) and ferrule. Synthetic handles shall be dropped twice from a height of three meters (10 feet) onto a concrete surface and examined for compliance with 3.1.4.

4.4.4 **Straightness**. The assembled brush head shall be placed so that the bristle is immersed in 60 degree C (140 degree F) water for five (5) minutes. Excess water shall be shaken out. The brush shall be hung with bristle down to dry completely. The flare shall be measured by aligning the ferrule portion of the brush as shown in figure 1. Flare measurement is limited to the outline of the main body of the brush. Stray bristles shall be disregarded.

4.4.5 **Tipping and flagging of filament**. The depth of tipping and flagging shall be measured with a ruler and a 7-power magnifying glass.

5. PREPARATION FOR DELIVERY

5.1 **Preservation, packaging, packing, and marking**. The preservation, packaging, packing, and marking shall be in accordance with PPP-B-1880. The level of packaging and packing shall be A, B, C, or otherwise specified in the contract or order (see 6.2). The marking shall be civil, military, or as specified (see 6.2).

6. Notes

6.1. **Intended Use**. Brushes made with natural hog bristle (Type I) are recommended for application of solvent borne paints and coatings. Brushes made with synthetic filaments (Type II) are recommended for application of water borne paints and coatings.

6.2 **Ordering data**. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type, grade, and size required (see 1).
(c) Levels of packaging and packing required (see 5.1).
(d) Marking required (see 5.1).

6.3 Definitions.

6.3.1 **Flagging**. Flagging is a mechanical process to split synthetic filament ends to simulate the natural flag of hog bristle.

6.3.2. **Tipping**. Tipping is a process for grinding or sanding synthetic filament tips to produce a fine, soft, even tip.

PREPARING ACTIVITY

GSA - FSS

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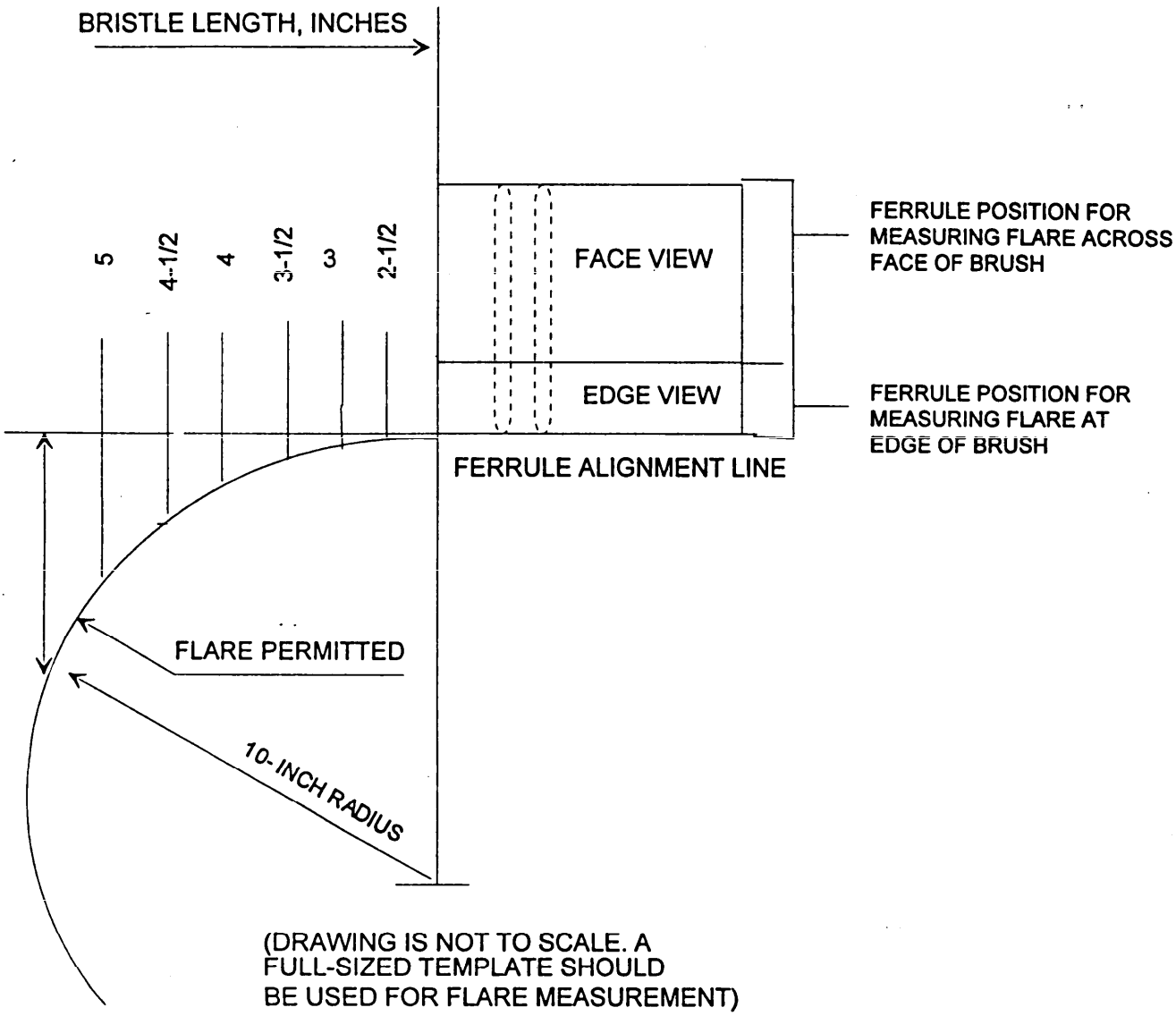


Figure 1. Flare Measurement

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